

task1

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0.1 Deakin University

0.1.1 SIG731- Data Wrangling

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0.2 Task1 discription

The main source of income for the Australian Government is income tax, which is collected by the Australian Taxation Office (ATO) and applied to the taxable income of individuals. Suppose your employer pays your salary on a weekly basis. They will deduct the weekly withholding tax from your salary and send it to the ATO. At the end of the financial year, you can lodge a tax return to calculate your tax liability. Often, your employer will have already paid a little too much tax to the ATO, making you eligible for a tax return.

0.2.1 Question Details:

In this exercise, you will be asked to complete the following questions:

1. Generate 10 individuals' weekly salaries using the given distribution.
2. Calculate their yearly income tax based on the Australian resident tax rates for 2023–24.
3. Calculate their weekly withholding tax using the withholding tax formula provided by the ATO.
4. Calculate each individual's tax return.
5. Calculate the employer's weekly before-tax superannuation contribution.
6. Repeat Subtasks 2, 3, 4 to recalculate the income tax, withholding tax, and tax return.

```
[15]: import random
import numpy as np
```

```
[16]: weeklySalary = np.random.normal(1431, 527, 10)
print(f"Weekly Salary:${weeklySalary}")
```

```
Weekly Salary:[1776.7348372  1714.00606666 2052.6364367  1613.30182625
2025.55002446
1016.35437622 1492.1302313  1760.48984507 1076.44230426 2143.88208013]
```

Without using Numpy Library and Generating 10 Individual Weekly Salary

0.3 Q1. Generate 10 individuals' weekly salaries using the given distribution.

```
[17]: def generate_weekly_salaries(mean, std_dev, num_individuals):
    salaries = []
    for _ in range(num_individuals):
        # Use random.gauss to generate salaries based on mean and std deviation
        salary = random.gauss(mean, std_dev)
        salaries.append(round(salary, 2)) # Round to 2 decimal places
    return salaries

# Given Parameters
mean_salary = 1431
std_dev_salary = 527
num_individuals = 10

# weekly salaries
weekly_salaries = generate_weekly_salaries(mean_salary, std_dev_salary,
↪num_individuals)

# List of salaries
print("Generated Weekly Salaries:", weekly_salaries)
```

Generated Weekly Salaries: [1018.37, 1210.73, 589.45, 416.07, 1751.24, 685.88, 1303.35, 2115.85, 1037.81, 1935.41]

generate_weekly_salaries(arg1,arg2,arg3) method takes three argument mean, standard deviation

salary = random.gauss(mean, std_dev) its return the random gauss salary based on Mean & std dev

Given Parameters

```
-----
mean_salary = 1431
std_dev_salary = 527
num_individuals = 10
```

0.4 Q2: Calculate annual income tax for each individual based on their weekly salary

```
[18]: def calculate_annual_income_tax(weekly_salaries):
    income_taxes = []
    for weekly_salary in weekly_salaries:
        annual_income = weekly_salary * 52
        if annual_income <= 18200:
            tax = 0
        elif annual_income <= 45000:
            tax = 0.19 * (annual_income - 18200)
```

```

    elif annual_income <= 120000:
        tax = 5092 + 0.325 * (annual_income - 45000)
    elif annual_income <= 180000:
        tax = 29467 + 0.37 * (annual_income - 120000)
    else:
        tax = 51667 + 0.45 * (annual_income - 180000)
    income_taxes.append(round(tax, 2))
return income_taxes

# Calculate income tax for each individual
income_taxes = calculate_annual_income_tax(weekly_salaries)

# List of Income Tax
print("Annual Income Taxes:", income_taxes)

```

Annual Income Taxes: [7677.45, 10928.34, 2365.77, 652.77, 20062.96, 3318.49, 12493.61, 26224.86, 8005.99, 23175.43]

Based on Australian task brackets. This calculate_annual_income_tax() method Return Annual income tax for indivisual salary.

Annual Tax slave are calculated based on Person Income

0.4.1 Q3: Calculate weekly withholding tax for each individual based on their weekly salary

```

[19]: def calculate_weekly_withholding_tax(weekly_salaries):
    withholding_taxes = []
    for weekly_salary in weekly_salaries:
        x = weekly_salary + 0.99 # Weekly salary plus 99 cents
        if x < 359:
            tax = 0
        elif x < 438:
            tax = 0.19 * x - 68.3462
        elif x < 548:
            tax = 0.29 * x - 112.1942
        elif x < 721:
            tax = 0.21 * x - 68.3465
        elif x < 865:
            tax = 0.219 * x - 74.8369
        elif x < 1282:
            tax = 0.3477 * x - 186.2119
        elif x < 2307:
            tax = 0.345 * x - 182.7504
        elif x < 3461:
            tax = 0.39 * x - 286.5965
        else:
            tax = 0.47 * x - 563.5196

```

```

        withholding_taxes.append(round(tax, 2))
    return withholding_taxes

# Calculate withholding tax for each individual
withholding_taxes = calculate_weekly_withholding_tax(weekly_salaries)

# Output the withholding tax values
print("Weekly Withholding Taxes:", withholding_taxes)

```

Weekly Withholding Taxes: [168.22, 235.1, 55.65, 10.9, 421.77, 75.9, 267.25, 547.56, 174.98, 485.31]

This Method calculate_weekly_withholding_tax()

takes weekly salaries and calculate the weekly without holding tax based on Given Condition parameter,

without holding tax amount of individual person

0.5 Q4: Calculate the annual eligible tax refund for each individual

```

[20]: def calculate_tax_return(income_taxes, withholding_taxes):
    tax_returns = []
    for income_tax, weekly_withholding_tax in zip(income_taxes,
    ↳withholding_taxes):
        annual_withholding_tax = weekly_withholding_tax * 52
        tax_return = annual_withholding_tax - income_tax
        tax_returns.append(round(tax_return, 2))
    return tax_returns

# Calculate tax returns for each individual
tax_returns = calculate_tax_return(income_taxes, withholding_taxes)

# Output List of Annual Tax for each Individual
print("Annual Tax Returns:", tax_returns)
print("-----"*4)

# Print detailed information for each individual
for i, (weekly_salary, weekly_withholding_tax, income_tax, tax_return) in
↳enumerate(zip(weekly_salaries, withholding_taxes, income_taxes,
↳tax_returns), start=1):
    weekly_income = weekly_salary - weekly_withholding_tax
    print(f"## Person {i}")
    print(f"Weekly salary: ${weekly_salary:.2f}")
    print(f"Weekly withholding tax: ${weekly_withholding_tax:.2f}")
    print(f"Weekly income: ${weekly_income:.2f}")

```

```
print(f"Income tax: ${income_tax:.2f}")
print(f"Tax return: ${tax_return:.2f}")
print("-----"*3)
```

Annual Tax Returns: [1069.99, 1296.86, 528.03, -85.97, 1869.08, 628.31, 1403.39, 2248.26, 1092.97, 2060.69]

Person 1

Weekly salary: \$1018.37
Weekly withholding tax: \$168.22
Weekly income: \$850.15
Income tax: \$7677.45
Tax return: \$1069.99

Person 2

Weekly salary: \$1210.73
Weekly withholding tax: \$235.10
Weekly income: \$975.63
Income tax: \$10928.34
Tax return: \$1296.86

Person 3

Weekly salary: \$589.45
Weekly withholding tax: \$55.65
Weekly income: \$533.80
Income tax: \$2365.77
Tax return: \$528.03

Person 4

Weekly salary: \$416.07
Weekly withholding tax: \$10.90
Weekly income: \$405.17
Income tax: \$652.77
Tax return: \$-85.97

Person 5

Weekly salary: \$1751.24
Weekly withholding tax: \$421.77
Weekly income: \$1329.47
Income tax: \$20062.96
Tax return: \$1869.08

Person 6

Weekly salary: \$685.88
Weekly withholding tax: \$75.90
Weekly income: \$609.98

Income tax: \$3318.49
Tax return: \$628.31

Person 7
Weekly salary: \$1303.35
Weekly withholding tax: \$267.25
Weekly income: \$1036.10
Income tax: \$12493.61
Tax return: \$1403.39

Person 8
Weekly salary: \$2115.85
Weekly withholding tax: \$547.56
Weekly income: \$1568.29
Income tax: \$26224.86
Tax return: \$2248.26

Person 9
Weekly salary: \$1037.81
Weekly withholding tax: \$174.98
Weekly income: \$862.83
Income tax: \$8005.99
Tax return: \$1092.97

Person 10
Weekly salary: \$1935.41
Weekly withholding tax: \$485.31
Weekly income: \$1450.10
Income tax: \$23175.43
Tax return: \$2060.69

calculate_tax_return() method Calculate the annual eligible tax for each individual based on Given condition.

it's take the Two Parameter Argument .

- income_takes: List of Annual Income Taxes for each Individual.
- with_holding_taxes: List of weekly with holding taxes for each individual

0.5.1 Q4. Calculate each individual's tax return.(Improve using calculate_tax_return_and_print() method)

```
[21]: def calculate_tax_return_and_print(weekly_salaries, withholding_taxes, income_taxes):  
    tax_returns = []  
    for i in range(len(weekly_salaries)):  
        weekly_salary = weekly_salaries[i]  
        annual_withholding_tax = withholding_taxes[i] * 52
```

```

        income_tax = income_taxes[i]
        tax_return = annual_withholding_tax - income_tax
        weekly_income_after_tax = weekly_salary - withholding_taxes[i]

        tax_returns.append(tax_return)

        # Print details for the individual
        print(f"## Person {i + 1} weekly salary \n ${weekly_salary:.2f}")
        print(f"weekly withholding tax ${withholding_taxes[i]:.2f}")
        print(f"weekly income\n${weekly_income_after_tax:.2f}")
        print(f"income tax ${income_tax:.2f} tax return ${tax_return:.2f}.\n")
        print("-----"*3)

    return tax_returns

#Calling Method
calculate_tax_return_and_print(weekly_salaries, withholding_taxes, income_taxes)

```

```

## Person 1 weekly salary
$1018.37
weekly withholding tax $168.22
weekly income
$850.15
income tax $7677.45 tax return $1069.99.

```

```

-----
## Person 2 weekly salary
$1210.73
weekly withholding tax $235.10
weekly income
$975.63
income tax $10928.34 tax return $1296.86.

```

```

-----
## Person 3 weekly salary
$589.45
weekly withholding tax $55.65
weekly income
$533.80
income tax $2365.77 tax return $528.03.

```

```

-----
## Person 4 weekly salary
$416.07
weekly withholding tax $10.90
weekly income
$405.17

```

income tax \$652.77 tax return \$-85.97.

Person 5 weekly salary

\$1751.24

weekly withholding tax \$421.77

weekly income

\$1329.47

income tax \$20062.96 tax return \$1869.08.

Person 6 weekly salary

\$685.88

weekly withholding tax \$75.90

weekly income

\$609.98

income tax \$3318.49 tax return \$628.31.

Person 7 weekly salary

\$1303.35

weekly withholding tax \$267.25

weekly income

\$1036.10

income tax \$12493.61 tax return \$1403.39.

Person 8 weekly salary

\$2115.85

weekly withholding tax \$547.56

weekly income

\$1568.29

income tax \$26224.86 tax return \$2248.26.

Person 9 weekly salary

\$1037.81

weekly withholding tax \$174.98

weekly income

\$862.83

income tax \$8005.99 tax return \$1092.97.

Person 10 weekly salary

\$1935.41

weekly withholding tax \$485.31

weekly income

\$1450.10

income tax \$23175.43 tax return \$2060.69.

```
[21]: [1069.9900000000007,
       1296.8599999999988,
       528.0299999999997,
       -85.96999999999991,
       1869.0800000000017,
       628.3100000000004,
       1403.3899999999994,
       2248.2599999999948,
       1092.9699999999993,
       2060.6899999999987]
```

`calculate_tax_return_and_print()` method

Calculate tax return for each person as a print formatted. and it's take three Argument Params

Parameters: - `weekly_salaries`: List of weekly salaries. - `withholding_taxes`: List of weekly withholding taxes. - `income_taxes`: List of annual income taxes.

0.5.2 Q5. Calculate the employer's weekly before-tax superannuation contribution.

```
[22]: #Calculate weekly before-tax superannuation contributions
def calculate_super_contributions(weekly_salaries, contribution_rate=0.11):
    super_contributions = []
    for salary in weekly_salaries:
        contribution = salary * contribution_rate
        super_contributions.append(contribution)
    return super_contributions

# Calculate and print superannuation contributions
super_contributions = calculate_super_contributions(weekly_salaries)
for i, contribution in enumerate(super_contributions, start=1):
    print(f"## Person {i} weekly superannuation contribution\n${contribution:.
    ↵2f}")
```

```
## Person 1 weekly superannuation contribution
$112.02
## Person 2 weekly superannuation contribution
$133.18
## Person 3 weekly superannuation contribution
$64.84
## Person 4 weekly superannuation contribution
$45.77
## Person 5 weekly superannuation contribution
$192.64
## Person 6 weekly superannuation contribution
```

```

$75.45
## Person 7 weekly superannuation contribution
$143.37
## Person 8 weekly superannuation contribution
$232.74
## Person 9 weekly superannuation contribution
$114.16
## Person 10 weekly superannuation contribution
$212.90

```

`calculate_super_contributions()` this method calculate superannual tax contribution weekly before tax

Then Print out the lis of weekly supper annula contribution

0.5.3 Q6. Repeat Subtasks 2, 3, 4 to recalculate the income tax, withholding tax, and tax return.

```

[23]: #Calculate the Yearly Income tax
def calculate_yearly_income_tax(taxable_income):
    if taxable_income <= 18200:
        return 0
    elif taxable_income <= 45000:
        return 0.19 * (taxable_income - 18200)
    elif taxable_income <= 120000:
        return 5092 + 0.325 * (taxable_income - 45000)
    elif taxable_income <= 180000:
        return 29467 + 0.37 * (taxable_income - 120000)
    else:
        return 51667 + 0.45 * (taxable_income - 180000)

#Calculating taxholding base on weekly salary
def calculate_withholding_tax(weekly_salary):
    if weekly_salary < 359:
        return 0
    elif weekly_salary < 438:
        return 0.19 * (weekly_salary + 0.99) - 68.3462
    elif weekly_salary < 548:
        return 0.29 * (weekly_salary + 0.99) - 112.1942
    elif weekly_salary < 721:
        return 0.21 * (weekly_salary + 0.99) - 68.3465
    elif weekly_salary < 865:
        return 0.219 * (weekly_salary + 0.99) - 74.8369
    elif weekly_salary < 1282:
        return 0.3477 * (weekly_salary + 0.99) - 186.2119
    elif weekly_salary < 2307:
        return 0.345 * (weekly_salary + 0.99) - 182.7504
    elif weekly_salary < 3461:

```

```

        return 0.39 * (weekly_salary + 0.99) - 286.5965
    else:
        return 0.47 * (weekly_salary + 0.99) - 563.5196

def calculate_tax_return(annual_income_tax, annual_withholding_tax):
    return annual_withholding_tax - annual_income_tax

def calculate_superannuation(weekly_salary, rate=0.11):
    return weekly_salary * rate

# List of superannuation contributions
super_contributions = [calculate_superannuation(salary) for salary in
    ↪weekly_salaries]

# Recalculate income tax, withholding tax, and tax return considering
    ↪superannuation as a deduction
results = []
for i, weekly_salary in enumerate(weekly_salaries):
    super_contribution = super_contributions[i]
    taxable_income = (weekly_salary - super_contribution) * 52
    annual_income_tax = calculate_yearly_income_tax(taxable_income)
    weekly_withholding_tax = calculate_withholding_tax(weekly_salary -
    ↪super_contribution)
    annual_withholding_tax = weekly_withholding_tax * 52
    tax_return = calculate_tax_return(annual_income_tax, annual_withholding_tax)

    results.append({
        "Weekly Salary": weekly_salary,
        "Weekly Superannuation": super_contribution,
        "Weekly Taxable Income": weekly_salary - super_contribution,
        "Annual Income Tax": annual_income_tax,
        "Weekly Withholding Tax": weekly_withholding_tax,
        "Tax Return": tax_return,
        "Weekly Income After Tax": weekly_salary - weekly_withholding_tax -
    ↪super_contribution
    })

#Output of Result
for i, result in enumerate(results, start=1):
    print(f"## Person {i}")
    print(f"Weekly salary (including superannuation): ${result['Weekly Salary']:
    ↪.2f}")
    print(f"Weekly superannuation contribution: ${result['Weekly
    ↪Superannuation']:.2f}")

```

```

print(f"Weekly taxable income: ${result['Weekly Taxable Income']:.2f}")
print(f"Weekly withholding tax: ${result['Weekly Withholding Tax']:.2f}")
print(f"Weekly income after tax: ${result['Weekly Income After Tax']:.2f}")
print(f"Annual income tax: ${result['Annual Income Tax']:.2f}")
print(f"Tax return: ${result['Tax Return']:.2f}")
print("-----"*3)

```

Person 1

Weekly salary (including superannuation): \$1018.37
 Weekly superannuation contribution: \$112.02
 Weekly taxable income: \$906.35
 Weekly withholding tax: \$129.27
 Weekly income after tax: \$777.08
 Annual income tax: \$5784.30
 Tax return: \$937.74

Person 2

Weekly salary (including superannuation): \$1210.73
 Weekly superannuation contribution: \$133.18
 Weekly taxable income: \$1077.55
 Weekly withholding tax: \$188.80
 Weekly income after tax: \$888.75
 Annual income tax: \$8677.59
 Tax return: \$1139.82

Person 3

Weekly salary (including superannuation): \$589.45
 Weekly superannuation contribution: \$64.84
 Weekly taxable income: \$524.61
 Weekly withholding tax: \$40.23
 Weekly income after tax: \$484.38
 Annual income tax: \$1725.15
 Tax return: \$366.81

Person 4

Weekly salary (including superannuation): \$416.07
 Weekly superannuation contribution: \$45.77
 Weekly taxable income: \$370.30
 Weekly withholding tax: \$2.20
 Weekly income after tax: \$368.10
 Annual income tax: \$200.59
 Tax return: \$-86.22

Person 5

Weekly salary (including superannuation): \$1751.24
 Weekly superannuation contribution: \$192.64
 Weekly taxable income: \$1558.60

Weekly withholding tax: \$355.31
Weekly income after tax: \$1203.29
Annual income tax: \$16807.40
Tax return: \$1668.69

Person 6

Weekly salary (including superannuation): \$685.88
Weekly superannuation contribution: \$75.45
Weekly taxable income: \$610.43
Weekly withholding tax: \$60.05
Weekly income after tax: \$550.38
Annual income tax: \$2573.08
Tax return: \$549.64

Person 7

Weekly salary (including superannuation): \$1303.35
Weekly superannuation contribution: \$143.37
Weekly taxable income: \$1159.98
Weekly withholding tax: \$217.46
Weekly income after tax: \$942.52
Annual income tax: \$10070.69
Tax return: \$1237.12

Person 8

Weekly salary (including superannuation): \$2115.85
Weekly superannuation contribution: \$232.74
Weekly taxable income: \$1883.11
Weekly withholding tax: \$467.26
Weekly income after tax: \$1415.84
Annual income tax: \$22291.50
Tax return: \$2006.17

Person 9

Weekly salary (including superannuation): \$1037.81
Weekly superannuation contribution: \$114.16
Weekly taxable income: \$923.65
Weekly withholding tax: \$135.29
Weekly income after tax: \$788.37
Annual income tax: \$6076.70
Tax return: \$958.16

Person 10

Weekly salary (including superannuation): \$1935.41
Weekly superannuation contribution: \$212.90
Weekly taxable income: \$1722.51
Weekly withholding tax: \$411.86
Weekly income after tax: \$1310.66
Annual income tax: \$19577.50

Tax return: \$1839.16

calculate_yearly_income_tax(): this method calculate yearly income tax normal as per the Given task brackets

calculate_withholding_tax(): This method calculate withholding tax as per given condition based on the weekly salary

calculate_tax_return(): this method calculate Annual Income tax and Annual withholding tax

calculate_superannuation():this method calculate superannual salary contribution list for each individual

results Variable holds array of has list of

- Weekly Salary
- Weekly Superannuation
- Weekly Taxable Income
- Annual Income Tax
- Weekly Withholding Tax
- Tax Return
- Weekly Income After Tax

Finally All the result Context are print out

[]: